



DS 301 : Mathematical Foundations of Data Science

Course Code : DS 301

Course Name : Mathematical Foundations of Data Science

L-T-P-C : 3-1-0-4

Intended for : Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines

Prerequisite : IC110-Engineering Mathematics, IC111-Linear Algebra

Mutual Exclusion:

Approval: 55th BoA

Course Contents

- **Module 1:** Definition of metric spaces, Examples, Open sets, Closed sets, Dense sets, Compact sets, Connectedness, Closure and interior of the sets, Metric subspace. (10 Lectures)
- **Module 2:** Cauchy sequences, Convergent sequences, Complete metric space, , Continuous functions, Continuity of composite functions, Continuity and inverse image of open and closed sets. (10 Lectures)
- **Module 3:** Normed linear spaces, Linear subspaces of normed linear spaces, Banach spaces, Riesz lemma, Continuity of linear maps, Bounded linear maps, Norm equivalence. (10 Lectures)
- **Module 4:** Hilbert spaces, Cauchy -Schwarz inequality, Parallelogram law, Orthogonality, Pythagorean Theorem, Orthogonal projection, orthogonal complement and projection theorem, Orthonormal sets, Orthonormal basis, Gram-Schmidt process, Examples of orthonormal basis. (12 Lectures)

Textbooks:

1. Apostol, T., **Mathematical Analysis**, 2nd Edition, Narosa Publishers, 2002.
2. Limaye, B. V., **Functional Analysis**, 2nd Edition, New age international Publishers, 2009.
3. Dan Simovici, **Mathematical Analysis for Machine Learning and Data Mining**, World Scientific, 2018
4. Rudin, W., **Principles of Mathematical Analysis**, 3rd Edition, McGraw-Hill, 2013.

References:

1. Stein, E. M. and Shakarchi, M., **Real Analysis**, Princeton Lectures.
2. Tao, T, **Analysis I and II**, Trim, Hindustan book agency.
3. Kreyszig, E., **Introductory Functional Analysis with Applications**, Reprint, 2017
4. Naylor, A. C. and Sell, G. R., **Linear Operator Theory in Engineering and Science**.